

MEDIATEK

MT8668 Audio HW Interface User Guide

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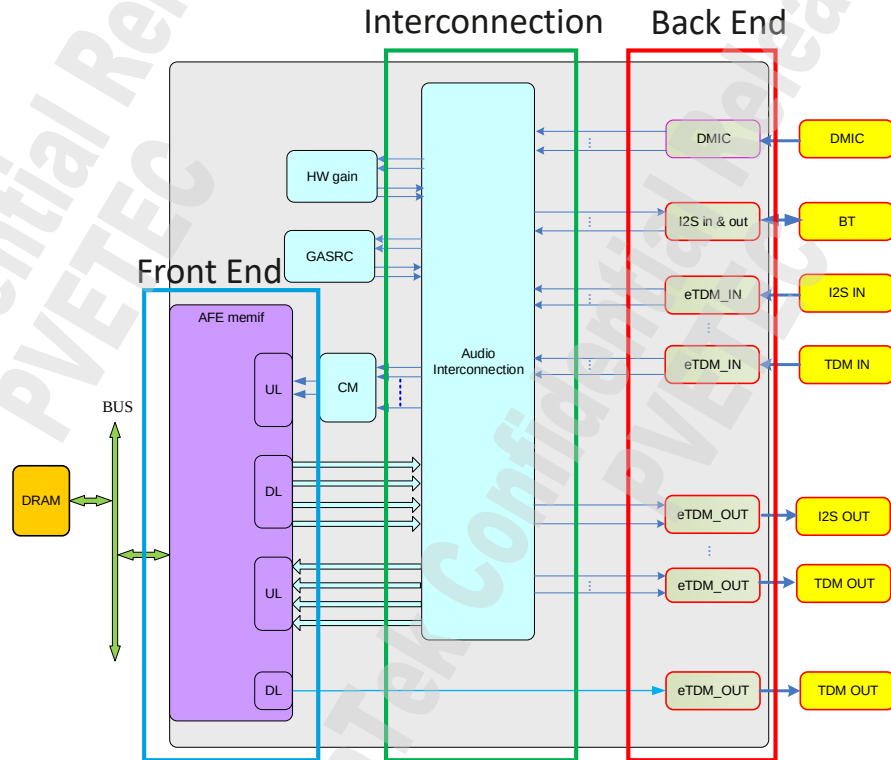
- ❑ MTK Audio HW Concept
- ❑ MT8668 AFE HW capability
- ❑ Audio HW Reference Design

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- ❑ **MTK Audio HW Concept**
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MTK Audio HW Concept (名词解释)

- AFE: Audio Front End HW
- AFE MEMIF(FE): PCM DMA, memory read/write
- Audio interconnection: connection fabric for audio sub module
 - ex: DL MEMIF -> Intercon -> I2S Out
- DAI (BE) : Digital Audio I/F, eTDM/I2S/DMIC.
 - Enhanced TDM, can be configured to I2S or TDM
- MISC
 - CM: Channel merge to combine echo reference and Mic input
 - GASRC: General audio sampling rate convertor(sample-based)
 - HW gain: digital gain process
 - Bus: SoC Top fabric bus



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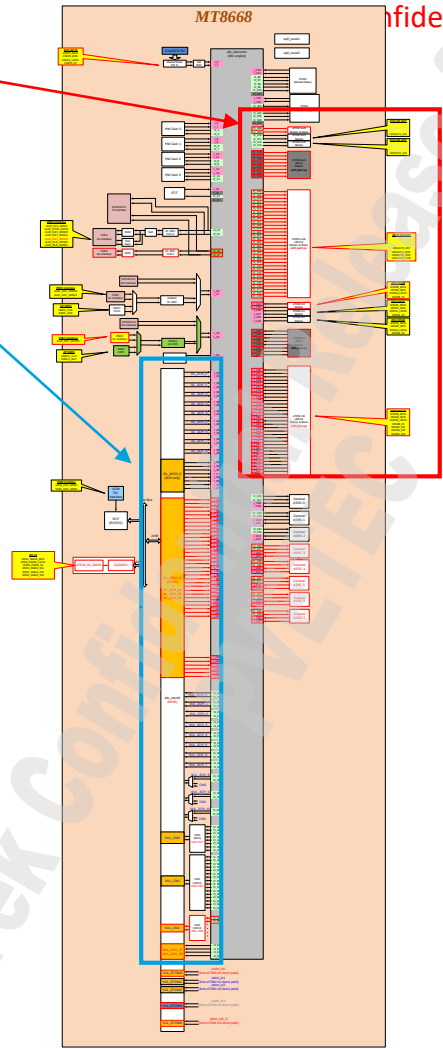
MT8668 AFE HW Capability

- 下行 **36ch**: 2组2ch I2S out, 1组32ch TDM out
- 上行 **48ch**: 4组2ch I2S in, 1组32ch TDM in, 1组8ch slave HS-I2S in

In/Out	Typical Usage	Mem IF(FE)	Audio IF(BE)
Max 36ch Output	2ch/48k/32b * 1 Data PIN	DL_2CH_X	I2SOUT1: 2ch
	2ch/48k/32b * 1 Data PIN	DL_2CH_X	I2SOUT2: 2ch
	8ch/48k/32b * 4 Data PINs 16ch/48k/32b * 2 Data PINs	DL_48CH_0(32ch)	I2SOUT6: 32ch TDM (Master & Slave)
Max 48ch Input	2ch/48k/32b * 1 Data PIN	VUL_2CH_X	FMI2S: 2ch (Master & Slave)
	2ch/48k/32b * 1 Data PIN	VUL_2CH_X	I2SIN0: 2ch
	2ch/48k/32b * 1 Data PIN	VUL_2CH_X	I2SIN1: 2ch
	2ch/48k/32b * 1 Data PIN	VUL_2CH_X	I2SIN2: 2ch
	8ch/48K/32b * 4 Data PINs 16ch/48k/32b * 2 Data PINs	VUL_CM2(32ch)	I2SIN6: 32ch TDM (Master & Slave)
	2ch/2048k/18b * 4 Data PINs	CQDMA	I2SIN_DMA0 (Slave only)

BE

FE



Audio Interfaces Capability

Output Audio IF	Audio Capability ^[*]	Support Slave?	独立 CLK?	PIN
I2SOUT1	2ch/192k/32b	No	No	O:I2SOUT1_DO
I2SOUT2	2ch/192k/32b	No	No	O:I2SOUT2_DO
I2SOUT6	16ch/48k/32b * 2 Data PINs 8ch/48k/32b * 4 Data PINs	Yes	No	O:I2SOUT6_DO O:I2SOUT6_DO1 O:I2SOUT6_DO2 O:I2SOUT6_DO3

Input Audio IF	Audio Capability ^[*]	Support Slave?	独立 CLK?	PIN
FMI2S	2ch/192k/32b	Yes	Yes	B0:FMI2SIN_BCK B0:FMI2SIN_LRCK I0:FMI2SIN_DI
I2SIN0	2ch/192k/32b	No	Yes	O:I2SIN0_BCK O:I2SIN0_LRCK I0:I2SIN0_DI
I2SIN1	2ch/192k/32b	No	No	O:I2SIN1_BCK O:I2SIN1_LRCK I0:I2SIN1_DI
I2SIN2	2ch/192k/32b	No	No	I0:I2SIN2_DI O:I2SIN2_BCK O:I2SIN2_LRCK
I2SIN6	16ch/48k/32b * 2 Data PINs 8ch/48k/32b * 4 Data PINs	Yes	No	O:I2SIN6_O_BCK O:I2SIN6_O_LRCK I0:I2SIN6_DI I0:I2SIN6_DI1 I0:I2SIN6_DI2 I0:I2SIN6_DI3
I2SIN_DMA0	2ch/2048k/18b * 4 Data PINs 2ch/192k/32b * 4 Data PINs	Yes (only slave)	Yes	I0:I2SIN_DMA0_BCK I0:I2SIN_DMA0_LRCK I0:I2SIN_DMA0_DI I0:I2SIN_DMA0_DI1 I0:I2SIN_DMA0_DI2 I0:I2SIN_DMA0_DI3

[1] MT8668 BCK 最大能力为 24.576MHz，单根 data pin 的传输数据带宽受此限制

[2] I2SOUTx和I2SINx 共用一组CLK PIN (例如I2SOUT0和I2SIN0)，需要LRCK/BCK配置保持一致

[3] I2SIN_DMA0 只支持 slave input，主要功能是为 SDR 设计的 HS-I2S，也能当做普通 I2S IN

Audio Interface Co-clock

- 如果考虑音频接口I2S/TDM的共用时钟，请按照如下规则确认，满足下面条件才有机会：
 - 音频格式 (sample rate/channel count/bit depth) 需要保持一致，BCK/LRCK(Sync Clock)要保持一致
 - Master/slave mode 要一致
 - 确认是否在可以共clock的 HW group

可以共clock的HW Group

I2SOUT1
I2SOUT2
I2SIN0
I2SIN1
I2SIN2

可以共clock的HW Group

I2SIN6
I2SOUT6

可以共clock的HW Group

FMI2S

可以共clock的HW Group

I2SIN_DMA0

Audio Interface Usage Description

In/Out	音频接口	使用说明
Max 36ch Output	I2SOUT1: 2ch	Channel count = 2ch
	I2SOUT2: 2ch	Channel count = 2ch
	I2SOUT6: 32ch TDM master & slave	BCK <= 24.576MHz, channel count <= 32ch 典型应用场景: 16ch/48k/32b * 2 Data PINs 或者 8ch/48K/32b * 4 Data PINs
Max 48ch Input	FMI2S: 2ch master & slave	Channel count = 2ch, 能够支持master和slave mode
	I2SIN0: 2ch	Channel count = 2ch
	I2SIN1: 2ch	Channel count = 2ch
	I2SIN2: 2ch	Channel count = 2ch
	I2SIN6: 32ch TDM master & slave	BCK <= 24.576MHz, channel count <= 32ch 典型应用场景: 16ch/48k/32b * 2 Data PINs 或者 8ch/48K/32b * 4 Data PINs
	I2SIN_DMA0: 8ch I2S slave(I2S2ch * 4)	BCK <= 73.728MHz, channel count = 2ch * 4 典型应用场景: 2ch/2048k/18b * 4 Data PINs 或者 2ch/48k/32b * 4 Data PINs

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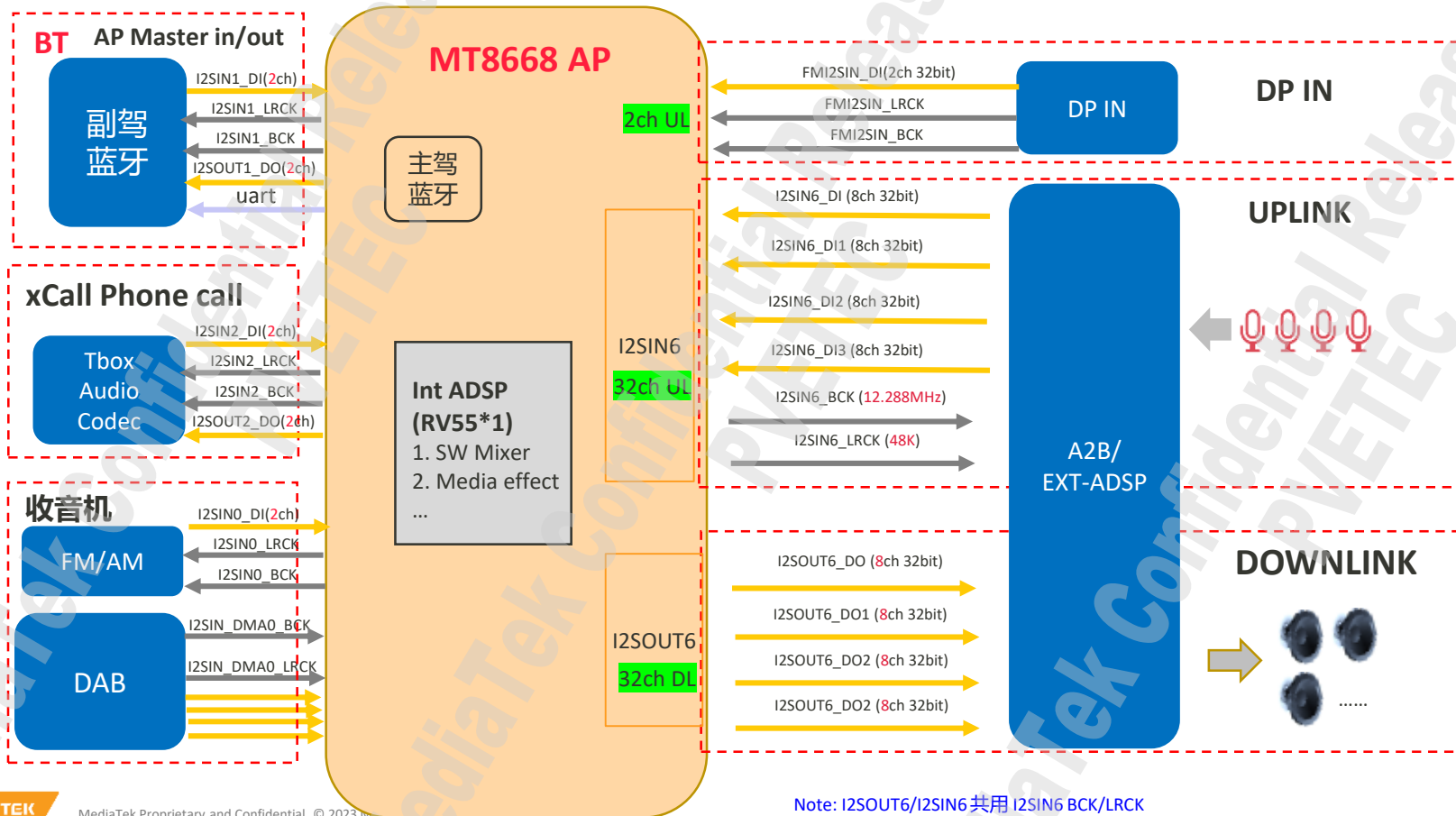
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MT8668 Audio Use Cases

Confidential C

Audio Clock →

Audio Data →



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Confidential C

Audio Clock →

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